ROLL NO: _____

MADHAV INSTITUTE OF TECHNOLOGY & SCIENCE, GWALIOR

(A Deemed to be University | Autonomous Institute Established in 1957 | NAAC 'A++' Accredited)

END-SEM EXAMINATION — DECEMBER / JUNE 2024

B.TECH. CSE ENGINEERING — SEMESTER 7

Time Allowed: 3 Hours **Maximum Marks: 70**

NOTE: 1. Attempt any FIVE questions selecting at least ONE question from each Unit.
2. All questions carry equal marks. Sub-questions carry marks indicated in brackets [ ].
3. Assume suitable missing data, if any, and state it clearly.
4. Use of scientific non-programmable calculator is permitted.

--- UNIT I ---

Q.1 Define AI and Turing Test. Explain PEAS description for Autonomous Driving Agent. [14 Marks]

[OR]

Q.1 (Alt) Compare Uninformed Search strategies: BFS, DFS, Uniform Cost Search, and Iterative Deepening. [14 Marks]

--- UNIT II ---

Q.2 Explain A* Search algorithm with admissibility and consistency of heuristics. Prove optimality. [14 Marks]

[OR]

Q.2 (Alt) Explain Hill Climbing, Simulated Annealing, and Genetic Algorithms for optimization. [14 Marks]

--- UNIT III ---

Q.3 Explain Adversarial Search: Minimax algorithm and Alpha-Beta Pruning with game tree example. [14 Marks]

[OR]

Q.3 (Alt) Explain Constraint Satisfaction Problems (CSP): Backtracking with Forward Checking and MRV heuristic. [14 Marks]

--- UNIT IV ---

Q.4 Explain First Order Predicate Logic (FOPL), Skolemization, and Resolution Refutation proof. [14 Marks]

[OR]

Q.4 (Alt) Describe Knowledge Representation: Semantic Networks, Conceptual Dependency, and Frames. [14 Marks]

--- UNIT V ---

Q.5 Explain Architecture of Expert Systems with Inference Engine and Knowledge Base. [14 Marks]

[OR]

Q.5 (Alt) Describe Bayesian Networks and Probabilistic Reasoning under Uncertainty with Bayes' Theorem. [14 Marks]

**** END OF QUESTION PAPER — MITS GWALIOR AUTONOMOUS EXAMINATION CELL ****